• Stakeholder Analysis and Conflict Resolution

Stakeholder Analysis

Method: Stakeholders were identified from the case study and the five interview statements (Principal, Lecturer, Head of Department, Student Representative and IT Officer). Each stakeholder was then rated for:

1. Influence, which is how much the stakeholders affect what can be built or how the system must operate.
2. Interest, which is how much the system affects the stakeholders

The analyzing criteria used is HIGH, MEDIUM and LOW

Table 4: Stakeholders and their interests

ID	Stakeholder	Role in the system	Main interests	Influence	Interest
1	Principal	Sponsor and final decision maker	A simple system for non-IT staff. Students must not see any mark until the Head of Department has checked it.	High	Medium
2	Head of Department (HoD)	Reviews, moderates and releases marks	Sees every mark in the department before release. Late-penalty rule applied consistently (10% per day for up to three days, then zero).	High	High
3	Lecturers	Create assignments, grant extensions,	Students get marks soon after marking. Ability to grant	Medium	High

ID	Stakeholder	Role in the system	Main interests	Influence	Interest
		mark work, write feedback	extensions for good reasons. Easy to use.		
4	Students (via student representative)	Submit work, resubmit, view released marks and feedback	Fix a mistake and resubmit before the deadline. Written feedback, not just a number. Late work not simply rejected.	Low	High
5	System administrator	Manages users, departments, courses, assigns lecturers, enrols students	Simple account and course management. Reliable system.	Medium	High
6	IT officer	Installs and maintains the system	Works offline on old computers. Passwords protected. Fast response with 800 students stored.	Medium	Medium
7	Parents and guardians (indirect)	Not users	Learners' personal data and marks kept private and accurate.	Low	Medium
8	Project supervisor	Reviews each phase gate	Requirements that are complete, consistent and verifiable.	High	Medium

Table 5: Influence and interest grid

	Lower interest	Higher interest
Higher influence	**Keep satisfied:** Principal, Supervisor	**Manage closely:** Head of Department
Lower influence	**Monitor:** Parents and guardians	**Keep informed:** Lecturers, Students, Administrator, IT officer

How the grid is used. The Principal and HoD decide the rules that most affect the design, so their statements carry the most weight when conflicts are resolved. Lecturers, students and the administrator use the system every day, so their needs shape the interface and workflow. Parents are not users, but they are the reason data protection and privacy are treated as requirements.

Ambiguities and Resolutions

Resolution rules. When stakeholders disagreed, or a statement was unclear, I applied these rules in order:

1. Rules stated in the case study come first.
2. Legal and ethical duties come next, such as protecting learners' data and treating them fairly.
3. The Principal's decision comes after that, as the person in charge of the school, while still meeting the other stakeholder's real need in another way wherever possible.

Table 6: Conflicts between stakeholders

ID	Conflict	Resolution	Reason
C1	**When students see marks:** The Principal says not until the HoD has checked them. The Lecturer wants students to see them as soon as they are marked.	Students see marks and feedback only after the HoD has moderated and released them. The lecturer's need for speed is met by notifying the HoD as soon as marking is finished.	The case study makes the HoD responsible for releasing marks, and the Principal has the final say.
C2	**Whether late work should be accepted or rejected:** The student representative says late work must not be rejected. The HoD says it loses marks and then gets zero.	Late work is always accepted and recorded. A penalty is applied, and after three days late the mark is zero. The lecturer can still write feedback.	Meets both needs: the work is never rejected, and the HoD's penalty rule still applies.
C3	**Extensions versus strict penalties:** The Lecturer wants to grant extensions. The HoD wants late work to be penalized.	A lecturer can grant an individual extension with a written reason, recorded in the system. The penalty is counted from the new deadline.	Fair to students with a good reason, and every extension is traceable so the HoD can review it.

ID	Conflict	Resolution	Reason
C4	**Simple system versus security and audit:** The Principal wants simplicity. The IT officer wants protected passwords, and the case study requires a log of every mark change.	Security works in the background (salted password hashes, audit log). The interface stays menu-driven and shows only the options for the user's role.	Users do not see the complexity, so both needs are met.

Table 7: Ambiguities and incomplete statements

ID	Ambiguity	Resolution
A1	"10% of the available mark for each day late": 10% of what, and what counts as a day?	The penalty is 10% of the assignment's maximum mark for each day or part of a day late, taken off the earned mark and never going below zero. More than three days late gives zero as last mark. Example: maximum 20, earned 16, two days late: penalty 4, final mark 12.
A2	Lateness is measured from which deadline?	From the effective deadline: the extended deadline if one was granted, otherwise the original deadline.
A3	"Resubmit before the deadline": what happens to earlier versions?	Resubmission is allowed only before the effective deadline. The latest version is the one marked, and earlier versions are kept in the submission history.
A4	What does the HoD actually do?	The HoD reviews every mark in their department, may change a mark with a reason (recorded in the audit log), then approves and releases it.
A5	"Respond quickly" is not measurable.	Replaced by measurable targets in the SRS, for example login within 2 seconds with 800 learner records stored.

ID	Ambiguity	Resolution
A6	"Single computer or local network": SQLite is a single file and is unreliable when several computers write to it over a network share.	**The system runs on one school computer.** The program and the SQLite database file are stored on that computer, and users work on it directly. The database file is never shared over the network. All database access goes through the data-access layer, so a networked version could be added later by changing only that layer. To be confirmed with the IT officer.
A7	When do students see feedback?	At the same time as the marks, after HoD release, to stay consistent with conflict 1.
A8	Until when can an extension be granted?	Until the submission has been marked. If a student has already submitted late, the penalty is recalculated against the new deadline.
A9	Can a mark change after it has been released?	Yes, but only the HoD can change a released mark, a reason is required, the change is recorded in the audit log (who, when, old value, new value, reason), and the student sees the updated mark.

Table 8: Business rules that result from the resolutions

These rules will become requirements in the SRS and will be used in the design, code and tests.

Rule	Statement	From
BR1	A student can see a mark and feedback only after the HoD has released it.	C1, A7
BR2	Late work is always accepted. Penalty = 10% of the maximum mark per day or part of a day late, up to three days. After three days the mark is zero.	C2, A1

Rule	Statement	From
BR3	Lateness is measured from the effective deadline, which includes any extension.	A2
BR4	An extension needs a written reason and is recorded. It can be granted until the submission is marked.	C3, A8
BR5	A student may resubmit only before the effective deadline. The latest version is marked and earlier versions are kept.	A3
BR6	The HoD can review, change (with a reason) and release all marks in their department.	A4
BR7	Every mark change is recorded with who, when, old value, new value and reason. Only the HoD can change a released mark.	C4, A9
BR8	The system runs on one school computer with a local SQLite database.	A6

• Software Requirements Specification

This SRS follows the structure of ISO/IEC/IEEE 29148 (introduction, overall description, specific requirements). It builds on the stakeholder analysis, the conflict resolutions (C1 to C4, A1 to A9) and the business rules (BR1 to BR8)

1. Introduction

➢ Purpose

This document states what the Learning Management System (LMS) must do and how well it must do it. It is the agreed basis for design, implementation and testing. Every requirement has a unique ID, a MoSCoW priority and a way to verify it.

➢ Scope

The LMS is a C++17 application for a secondary school. It manages courses, assignments, submissions, marking, moderation and release of marks for four user types: administrator, lecturer, Head of Department (HoD) and student.

In scope: login and role-based access, user, department and course management, assignments with deadlines and extensions, submission and resubmission, automatic late penalties, marking with feedback, HoD moderation and release, and an audit log of all mark changes.

Out of scope (from the brief): online payment, video conferencing, automated grading, plagiarism detection, and web or mobile interfaces.

➢ Definitions

Term	Meaning
HoD	Head of Department. Moderates and releases marks for their department.
Submission	A file or text a student hands in for an assignment. Each resubmission is a new version of the same submission.
Effective deadline	The extended deadline if the lecturer granted an extension, otherwise the original deadline.
Moderation	The HoD reviewing a mark, optionally changing it with a reason, and approving it.
Release	Making a moderated mark and feedback visible to the student.
Audit log	A permanent record of every mark change: who, when, old value, new value and reason.

Term	Meaning
MoSCoW	Must have, Should have, Could have and Won't have (this release).

2. Overall Description

➢ Product Perspective

The LMS is a stand-alone application. It runs offline on one school computer and stores all data in a local SQLite database. It does not connect to any other system.

➢ User Classes

User class	Main tasks	Technical skill
Administrator	Manage users, departments, courses, lecturer assignment and enrolment	Medium
Lecturer	Create assignments, grant extensions, mark, write feedback	Low (not IT experts)
Head of Department	Review, moderate and release marks	Low
Student	View courses and assignments, submit and resubmit, view released marks	Low to medium

➢ Operating Environment and Constraint

- **Technology:** C++17, SQLite, built with CMake. Data is accessed only through a data-access layer using parameterized queries.
- **One computer (BR8, A6):** the program and database file live on one school computer. The database file is never shared over a network.

- **Interface:** a menu-driven console application. Each role sees only the options it is allowed to use (C4).
- **Offline:** no internet connection is needed or used.
- **Hardware:** the school computers are old. Assumed minimum: dual-core processor, 4 GB RAM, 500 MB free disk space (to be confirmed with the IT officer).
- **Passwords:** stored only as salted hashes made with a recognized library.

➢ Assumptions and Dependencies

a) The school has about 800 students and fewer than 100 staff accounts
b) Users take turns on the single computer, so performance targets are measured for one active session with the full data set stored (this refines assumption A5).
c) The school provides the lists of users, courses and enrolments for the administrator to load.
d) A school day, for lateness, counts in 24-hour periods from the effective deadline.

3. Specific Requirements

➢ External Interfaces

- **User Interface:** text menus and prompts. Each screen shows the user's role and name. Error messages say what went wrong and how to fix it, without technical jargon.
- **Software interface:** SQLite database through the data-access layer only.
- **Hardware and communications interfaces:** none beyond the computer's keyboard, screen and local disk.

➢ Functional Requirements

Priority is MoSCoW. Source refers to stakeholder statements, conflict or ambiguity IDs, or business rules from table 8. Verification methods: T = test, D = demonstration, I = inspection.

Table 9: Functional requirements

ID	Requirement	Priority	Source	Verify
FR1	The system shall let a user log in with a username and password and shall then show only the menu options allowed for that user's role.	Must	Case study, C4	T
FR2	The administrator shall be able to: (a) create, edit and deactivate user accounts for all four roles; (b) create departments and courses; (c) assign lecturers to courses and a HoD to each department; (d) enrol students in courses.	Must	Case study	T
FR3	A lecturer shall be able to create an assignment for their own course with a title, description, deadline and maximum mark.	Must	Case study	T
FR4	A student shall be able to: (a) view their enrolled courses and the assignments in them, with deadlines; (b) view a mark and its feedback only after the HoD has released it. Before release, neither shall be visible.	Must	Case study, Principal, C1, A7, BR1	T
FR5	A student shall be able to submit work for an assignment. The system shall record the time and set the status to Submitted, or to Late if the effective deadline has passed. Late work shall never be rejected.	Must	C2, BR2	T
FR6	A student shall be able to resubmit before the effective deadline. The latest version shall be marked and earlier versions shall be kept. Resubmission after the effective deadline shall not be allowed.	Must	Student rep, A3, BR5	T
FR7	A lecturer shall be able to grant an individual student an extension, entering a written reason, until the submission has been marked. The	Must	C3, A8, BR4	T

ID	Requirement	Priority	Source	Verify
	extension and its reason shall be recorded.			
FR8	The system shall calculate the late penalty automatically: 10% of the assignment's maximum mark for each day or part of a day after the effective deadline, up to three days, and a mark of zero after that. The result shall never be below zero.	Must	HoD, A1, A2, BR2, BR3	T
FR9	A lecturer shall be able to mark a submission with a mark between zero and the maximum mark and write feedback text.	Must	Lecturer, student rep	T
FR10	Each submission shall follow the life cycle Submitted or Late, then Marked, then Moderated, then Released. The system shall reject any other change of status.	Must	Case study	T, I
FR11	A HoD shall be able to: (a) view all marks, with their status, for every course in their own department and no other department; (b) see a list of marked submissions awaiting moderation; (c) change a mark, entering a reason; (d) approve marks as moderated and release them to students. After release, only a HoD shall be able to change a mark, and the student shall then see the updated mark.	Must	Principal, HoD, C1, A4, A9, BR6, BR7	T
FR12	The system shall record every mark change in an audit log with who made it, when, the old value, the new value and the reason.	Must	Case study, BR7	T, I
FR13	The administrator and HoD shall be able to view the audit log and filter it by student, assignment or date.	Should	Case study	T

ID	Requirement	Priority	Source	Verify
FR14	Every user shall be able to change their own password.	Should	IT officer	T
FR15	A lecturer shall be able to export the marks of a course to a CSV file.	Could	Lecturer	T

➢ Non-Functional Requirements

Each non-functional requirement has a measurable target and a stated way to verify it.

Table 10: Non-functional requirements

ID	Category	Requirement	Priority	Verification
NFR1	Performance	Login shall complete within 2 seconds with 800 student records stored, on a computer with a dual-core processor and 4 GB RAM.	Must	Test: time 20 logins with 800 generated students. All 20 within 2 seconds.
NFR2	Performance	With 800 students, 50 courses, 500 assignments and 10,000 submissions stored, 95% of list and view operations shall finish within 2 seconds, and releasing marks for 100 submissions shall finish within 5 seconds.	Must	Test: timed runs on a generated data set.
NFR3	Security	Passwords shall be stored only as salted hashes made with a	Must	Inspection of the database file and code; test that

ID	Category	Requirement	Priority	Verification
		recognized library. No plain-text password shall appear in the database, logs or source code.		two users with the same password have different hashes.
NFR4	Security	All database access shall go through the data-access layer using parameterized queries.	Must	Inspection of the code; test with at least 10 SQL-injection style inputs, all of which are rejected or treated as plain text.
NFR5	Security	Every action shall be checked against the user's role. A user shall be unable to perform an action their role does not allow.	Must	Test: each role attempts every other role's actions. 0 successes.
NFR6	Reliability	Invalid input or a database error shall never crash the program. The user shall see an error message and return to a menu. A mark change and its audit entry shall be saved together or not at all.	Must	Test: at least 30 invalid inputs and simulated database errors, 0 crashes. A forced failure leaves no mark change without an audit entry.
NFR7	Availability	All features shall work with no internet connection.	Must	Test: run all system tests with the network disabled.
NFR8	Usability	A lecturer who has read only the user manual shall be able to create an assignment and	Should	Test with at least 3 volunteers, timed. Inspection of the menu structure.

ID	Category	Requirement	Priority	Verification
		mark a submission within 5 minutes, and every function shall be reachable within 3 menu levels.		
NFR 9	Portability	The system shall build with CMake and a C++17 compiler and run on Windows 10 or later and on Ubuntu 22.04 using no more than 512 MB RAM.	Should	Test: build and run the system tests on both. Measure memory use.
NFR 10	Maintainability	The business-logic layer shall be separate from the user interface and data access. Unit tests shall cover at least 70% of its lines, and the static analyzer shall report no high-severity findings.	Should	Coverage report, static analysis report, inspection of the design.
NFR 11	Recoverability	The administrator shall be able to back up the database to a file and restore it, and the restored data shall be identical to the original.	Should	Test: back up, change data, restore and compare record counts and contents.
NFR 12	Accessibility	All functions shall be usable with the keyboard alone, and no information shall be given by colour alone.	Could	Inspection and demonstration.

Summary and Verification Approach

Table 11: Requirements by priority

Priority	Functional	Non-functional	Total
Must	12	7	19
Should	2	4	6
Could	1	1	2
Won't (this release)	0	0	0
Total	**15**	**12**	**27**

All Must requirements will be implemented and tested. Each functional requirement will be linked to a use case, design elements, source files and test cases in the traceability matrix. Each non-functional requirement is verified with the measurement stated in Table 10 during the verification and validation phase.

➢ Business Rules

Business rules BR1 to BR8 (Table 8) apply to all requirements. Where a requirement quotes a rule, the rule is the authority.

➢ Data Requirements

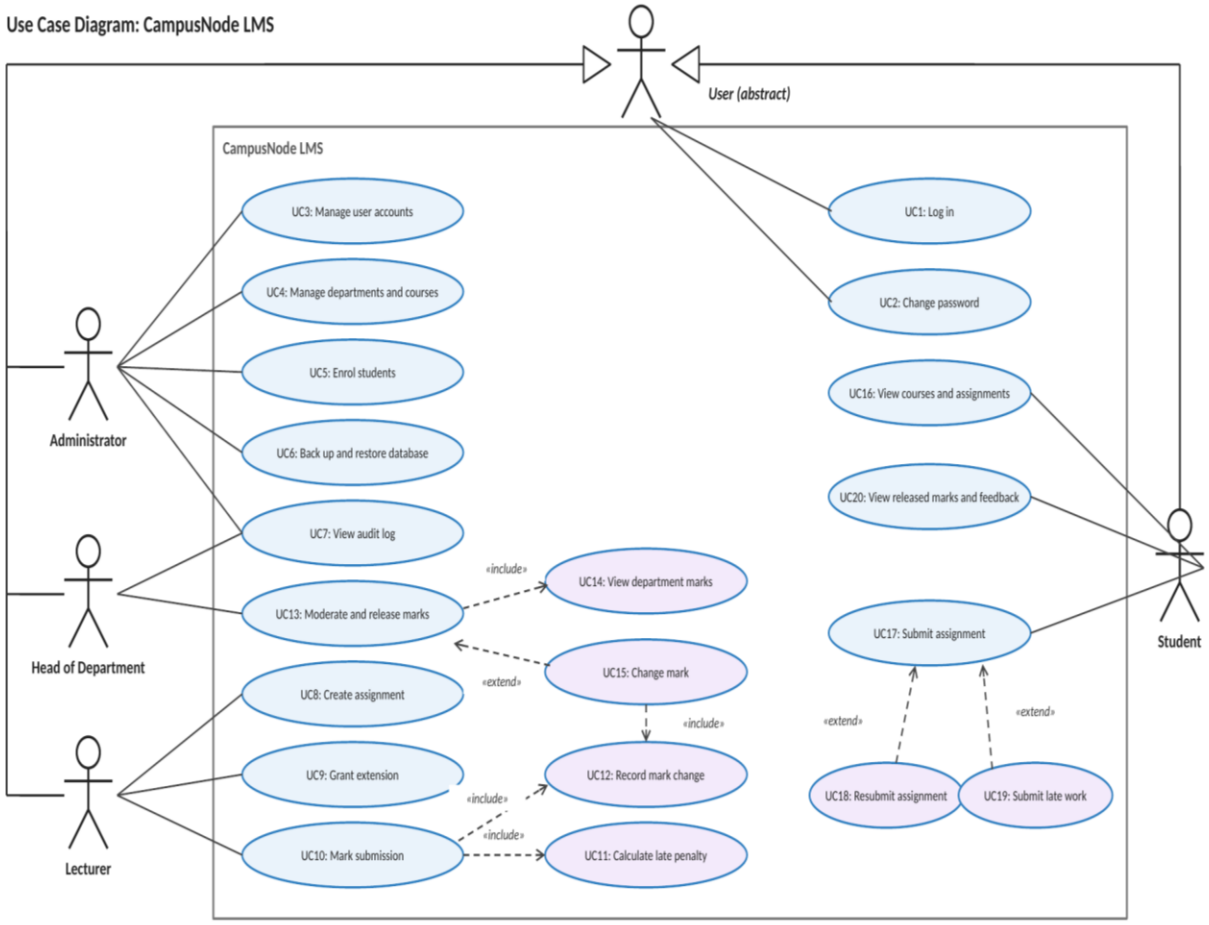
The system shall store at least the following, in a normalized database designed in Phase 3: users with role and password hash, departments, courses, enrolments, assignments, extensions, submissions and their versions, marks and feedback, and the audit log. Audit log records shall not be edited or deleted through the application.

• Use Case Model

1) Actors

Actor	Description
User (abstract)	Any person who logs in. It is never a real person, only the shared behaviour of all four roles.
Administrator	Manages the school's accounts and structure
Lecturer	Teaches courses and assesses work
Head of Department (HoD)	Responsible for the quality of marks in a department
Student	Learner enrolled in courses

2) Use Case Diagram



Generalization means that one thing is a more specific kind of another. Administrator, Lecturer, Head of Department and Student are all kinds of User, so they all share the User's behaviour. Logging in and changing a password belong to User, which avoids drawing the same two lines from every actor.

Include means a use case always carries out another use case as one of its steps. For example, marking a submission always calculates the late penalty and always records the change in the audit log, so Mark submission includes both. We made "Record mark change" its own use case because the audit rule applies to two use cases (marking and changing a mark). This way the rule is written once and cannot be forgotten. The arrow points from the main use case to the included one.

Extend means a use case adds optional behaviour that happens only under a condition. For example, Resubmit assignment happens only when the student already has a submission and the deadline has not passed. Submit late work happens only when the deadline has passed. Change mark happens only when the Head of Department disagrees with a mark. The arrow points from the optional use case to the one it extends

3) Use Case List

ID	Use case	Actor	Short description	Requirements
UC1	Log in	User	Enter username and password to reach the menu for the user's role	FR1, NFR3, NFR5
UC2	Change password	User	Change own password	FR14
UC3	Manage user accounts	Administrator	Create, edit and deactivate accounts for all four roles	FR2
UC4	Manage departments and courses	Administrator	Create departments and courses, assign lecturers and HoDs	FR2
UC5	Enrol students	Administrator	Enrol students in courses	FR2
UC6	Backup and restore database	Administrator	Save the database to a file and restore it	NFR11
UC7	View audit log	Administrator, HoD	View and filter the record of mark changes	FR13
UC8	Create assignment	Lecturer	Create an assignment with	FR3

			deadline and maximum mark	
UC9	Grant extension	Lecturer	Give one student a later deadline with a written reason	FR7
UC10	Mark submission	Lecturer	Enter a mark and feedback for a submission	FR8, FR9, FR10, FR11
UC11	Calculate late penalty	(included)	Apply 10% per day up to three days, then zero	FR8
UC12	Record mark change	(included)	Write who, when, old value, new value and reason to the audit log	FR12
UC13	Moderate and release marks	HoD	Review, approve and release department marks	FR10, FR11
UC14	View department marks	(included)	List all marks for the HoD's department	FR11
UC15	Change mark	(extends UC13)	Change a mark with a reason, including after release	FR11
UC16	View courses and assignments	Student	See enrolled courses, assignments and deadlines	FR4
UC17	Submit assignment	Student	Hand in work for an assignment	FR5, FR6, FR10
UC18	Resubmit assignment	(extends UC17)	Replace work before the effective deadline	FR6
UC19	Submit late work	(extends UC17)	Accept and flag work handed in after the deadline	FR5
UC20	View released marks and feedback	Student	See marks and feedback after release only	FR4

4) Fully Dressed Use Case Specification

UC17: Submit assignment

Field	Description
Use case ID and Name	UC17 Submit assignment
Primary Actor	Student
Goal	Hand in work for an assignment and receive confirmation, on time or late
Priority	Must
Requirements	FR5, FR6, FR10 (late penalty is applied at marking, FR8)
Extension points	Resubmit assignment (UC18), Submit late work (UC19)

Preconditions

1. The student is logged in (UC1) with the Student role.
2. The student is enrolled in the course that the assignment belongs to.
3. The assignment exists.
4. The student's submission for this assignment has not already been marked.

Trigger: The student chooses "Submit assignment" from the Student menu.

Main flow (first submission, before the deadline)

1. The system lists the assignments in the student's courses with their effective deadlines (UC16).
2. The student selects an assignment.
3. The system shows the title, description, maximum mark and effective deadline.
4. The student chooses "Submit work" and gives the file path or enters the text.
5. The system checks the input: it is not empty and not larger than the size limit (10 MB, to be confirmed).

6. The system compares the current time with the effective deadline. It is before the deadline, so the status is set to Submitted.
7. The system stores the work as version 1 with the date and time.
8. The system shows a confirmation with the time and status.

Alternative flows

- **A1: Late submission (UC19).** At step 6 the effective deadline has passed. The system sets the status to Late and warns the student that a penalty of 10% of the maximum mark applies for each day or part of a day late, up to three days, and that work more than three days late receives zero. The student confirms and the flow continues at step 7. If the student cancels, nothing is stored. Late work is never rejected (BR2).
- **A2: Resubmission (UC18).** At step 3 the student already has a submission and the effective deadline has not passed. The system shows the current version and asks the student to confirm replacing it. It stores the new work as the next version, keeps the earlier versions, and marks the latest version for marking (BR5). The flow continues at step 8.
- **A3: Extension granted.** If the lecturer has granted this student an extension (UC9), the effective deadline is the extended deadline. Steps 3 and 6 use it (BR3).

Exception flows

- **E1: Invalid work.** The input is empty, the file cannot be found or it is too large. The system shows a clear error message and returns to step 4. Nothing is stored.
- **E2: Resubmission after the deadline.** The system refuses with a message and keeps the existing submission (BR5).
- **E3: Already marked.** The submission has been marked, so it cannot be replaced. The system refuses with a message.
- **E4: Database error.** Saving fails. The system cancels the operation so nothing is half-saved, shows an error message and returns to the menu without crashing. The student can try again (NFR6).

Post Conditions

- **Success:** a new submission version is stored with its date, time and status (Submitted or Late). Earlier versions are kept. The lecturer can now see and mark the submission. The student has seen a confirmation.
- **Failure:** no data has changed.

Business rules

BR2 (late work accepted and penalised), BR3 (effective deadline), BR5 (resubmit only before the effective deadline, latest version marked, earlier versions kept). A day late means each 24-hour period or part of one after the effective deadline.

UC13: Moderate and release marks

Field	Description
Use case ID and name	UC13 Moderate and release marks
Primary actor	Head of Department
Goal	Check the marks in the department, change any that are wrong, approve them and release them to students
Priority	Must
Requirements	FR10, FR11, FR12
Includes	View department marks (UC14)
Extension points	Change mark (UC15), which includes Record mark change (UC12)

Preconditions

1. The HoD is logged in (UC1) with the HoD role.
2. The HoD is assigned to a department.
3. At least one submission in the department's courses has the status Marked.

Trigger: The HoD chooses "Moderate and release marks".

Main flow

1. The system lists the courses in the HoD's department with the number of marked submissions awaiting moderation (FR11).
2. The HoD selects a course and an assignment.
3. The system shows every submission (UC14): student, submission time, whether it was late, the mark before penalty, the penalty, the final mark, the feedback and the status.

4. The HoD reviews the marks and finds them correct.
5. The HoD chooses "Approve and release".
6. The system asks the HoD to confirm.
7. The system sets each selected submission to Moderated and then Released, saving all of them together as one operation.
8. The marks and feedback become visible to the students (BR1).
9. The system shows how many submissions were released.

Alternative flows

- **A1: Change a mark (UC15).** At step 4 the HoD disagrees with a mark. The HoD selects the submission and enters a new mark and a reason. The system checks that the mark is between zero and the maximum mark and that a reason was entered. It records the change in the audit log (UC12) with who, when, old value, new value and reason, updates the mark, and the flow returns to step 4.
- **A2: Approve without releasing.** The HoD chooses "Approve only" at step 5. The submissions become Moderated and stay hidden from students. They can be released later.
- **A3: Release some submissions only.** The HoD selects only some submissions at step 5. The others keep their current status.
- **A4: Change a mark after release.** The HoD opens a released submission and changes the mark (UC15). The audit log is updated and the student sees the new mark (A9, BR7).

Exception flows

- **E1: Nothing to moderate.** No submissions are in the marked status. The system shows a message and returns to the menu.
- **E2: Invalid mark.** The new mark is not a number or is outside the range. The system shows an error and asks again. The mark is unchanged.
- **E3: No reason.** The reason is empty. The system refuses the change and asks for a reason (BR7).
- **E4: Other department.** The HoD can only see and change marks in their own department. Other departments' courses are not listed and any attempt is refused (BR6, NFR5).
- **E5: Cancel.** The HoD cancels at step 6. Nothing changes.
- **E6: Database error.** The operation fails. The system cancels everything so that no mark is released without its audit entry, shows an error message and returns to the menu (NFR6).

Postconditions

- **Success:** approved submissions are Moderated or Released. Released marks and feedback are visible to students. Every mark change made during moderation has an audit log entry.

- **Failure:** no status or mark has changed.

Business rules

BR1 (students see marks only after release), BR6 (HoD reviews, changes with a reason and releases all marks in their own department), BR7 (every mark change is logged; only the HoD can change a released mark).

UC9: Grant extension

Field	**Description**
Use case ID and name	UC9 Grant extension
Primary actor	Lecturer
Goal	Give one student more time for one assignment, with a recorded reason
Priority	Must
Requirements	FR7 (affects FR5 and FR8)

Preconditions

1. The lecturer is logged in (UC1) with the Lecturer role.
2. The lecturer is assigned to the course.
3. The assignment exists and the student is enrolled in the course.
4. The student's submission for this assignment has not been marked.

Trigger: The lecturer chooses "Grant extension".

Main flow

1. The system lists the lecturer's courses. The lecturer selects a course and an assignment.
2. The system lists the enrolled students with their current effective deadlines.
3. The lecturer selects a student.
4. The system asks for a new deadline and a written reason.
5. The lecturer enters both.
6. The system checks that the new deadline is later than the current effective deadline and that the reason is not empty.
7. The system asks the lecturer to confirm.

8. The system stores the extension with the student, assignment, new deadline, reason, lecturer and time. The student's effective deadline is now the new deadline.
9. The system shows a confirmation.

Alternative flows

- **A1: Student already submitted late.** The student's submission has the status Late. After step 8 the system checks the submission time against the new deadline. If it is now on time, the status changes to Submitted. Any penalty is worked out from the new deadline when the work is marked (A8).
- **A2: Student already has an extension.** The new deadline must be later than the current effective deadline. The earlier extension stays in the record, and the newest one is used.
- **A3: Cancel.** The lecturer cancels at step 7. Nothing is stored.

Exception flows

- **E1: Invalid deadline.** The new deadline is not a valid date or is not later than the current effective deadline. The system shows an error and asks again.
- **E2: No reason.** The reason is empty. The system refuses and asks for a reason (BR4).
- **E3: Already marked.** The submission has been marked. The system refuses, because extensions are only possible until marking (A8).
- **E4: Wrong course.** Courses and students that do not belong to the lecturer's course are not listed (NFR5).
- **E5: Database error.** The system cancels the operation, shows an error message and returns to the menu without crashing (NFR6).

Postconditions

- **Success:** the extension is stored with its reason, and the student's effective deadline is updated and used for every later lateness check.
- **Failure:** no data has changed.

Business rules

BR3 (lateness is measured from the effective deadline), BR4 (an extension needs a written reason and can be granted until the submission is marked).